Development of an Algorithm for the Simultaneous Retrieval of Cloud-Top Height and Cloud Optical Thickness Combining Radiative Transfer and Multisource Satellite Information From O_4 Hyperspectral Measurements

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***Abstract*— Remote sensing of cloud properties based on multispectral or hyperspectral observations from satellites is important for Earth radiation budget and climate change studies. Currently, most retrieval algorithms for the hyperspectral measurements are developed based on the O_2-A band to derive cloud optical thickness (COT) and cloud top height (CTH) via the optimal estimation theory. Nevertheless, there are few studies on the retrieval of COT and CTH using the O_4 band, where the direct computation of slant column density (SCD) and spectral information in the blue band provide a faster yet flexible inversion strategy. In this study, we develop a novel cloud retrieval algorithm based on neural networks using the O_4 (CRANN-O4) band for the simultaneous derivation of COT and CTH. CRANN-O4 employs a transfer learning strategy that combines the radiative transfer model (RTM) and multisource satellite data, for which the deep neural network module is pretrained based on the simulation data from RTM to enhance its adaptability and interpretability, following a fine-tuning scheme using multisource satellite data. To evaluate the CRANN-O4 performance, we apply CRANN-O4 to TROPOMI and make an intercomparison with its official products, which is generated based on the O_2-A band. The results indicate that the CRANN-O4-derived spatial distributions of COT and CTH are generally similar to the official TROPOMI cloud product but are more consistent with the SNPP-VIIRS cloud product. The RMSEs of COT and CTH derived by CRANN-O4 are approximately 15.88 and 2.33 km, respectively, while those of the TROPOMI cloud product are 20.85 and 3.00 km, respectively. In addition, the validation of CRANN-O4-derived CTH using CALIOP measurements demonstrates better agreement than that of the TROPOMI official cloud product, with RMSE decreasing from 2.7 to 2.2 km. The methodology presented in this study provides innovative insight into cloud parameter retrieval for hyperspectral instruments with O_4 channels, such as FY-3F/OMS.**

***Index Terms*— Cloud optical thickness (COT), cloud-top height (CTH), O_4 absorption, transfer learning.**

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I. Introduction

THE IPCC AR6 report indicates that clouds are one of the most unpredictable factors affecting climate change [1], which is mainly reflected in the radiative effects of clouds and cloud-aerosol interactions [2], [4], [5], [6]. In addition to global impacts, clouds also have important implications for satellite-based observations of the atmosphere, especially for trace gas inversion [7], [8]. To correct for the effect of clouds on trace gas inversion, clouds are described through parameters such as cloud-top height (CTH), equivalent radiative cloud fraction (CF), and cloud optical thickness (COT) [7], [8], [9].

Current satellites provide the capability to retrieve cloud parameters using both multispectral and hyperspectral instruments. For COT retrieval with multispectral instruments, the classical approach using 0.75 and 2.16 μm two-channel simultaneous retrieval of COT and cloud effective radius is still widely utilized in many meteorological satellites [10], [11], [12], [13], [14]. For CTH retrieval with multispectral instruments, one of the most common classical methods is the CO_2-slicing method [15], [16], which is less sensitive to low clouds [17]. Multispectral-based cloud retrieval methods have been coupled with instruments such as moderate resolution imaging spectroradiometer (MODIS), visible infrared imaging radiometer suite (VIIRS), and various other instruments,

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enabling continuous global cloud observations and providing extensive data for climate change research [15], [16].

Atmospheric hyperspectral instruments can offer valuable cloud information, supporting both global energy balance studies and trace gas inversion [18], [19], [20]. To retrieve cloud parameters such as CTH, CF, and COT for hyperspectral instruments, O_2-A absorption band which is at 760–770 nm is often used in the retrieval algorithm, and most of these retrieval algorithms assume the cloud as a Lambertian body and employ the independent pixel approximation (IPA) to simplify radiative transfer modeling. For example, the fast retrieval scheme for clouds from the oxygen a band (FRESCO) algorithm retrieves the CF and cloud pressure using the O_2-A band and relies on Lambertian cloud assumptions with a fixed cloud albedo of 0.8, and the FRESCO algorithm has been successfully applied to global ozone monitoring experiment (GOME) observations [21], [22], [23]. Similarly, retrieval of cloud information using neural networks-clouds as reflecting boundaries (ROCINN-CRB) is the O_2-A band-based algorithm used to retrieve the CTH and cloud albedo based on Lambertian cloud assumptions, which is successfully applied to TROPOMI observations. In addition, OCRA/ROCINN-clouds-as-layers (CALs) is also the official algorithm applied to TROPOMI, which treats the cloud as a homogeneous layer with scattering properties rather than as a Lambertian body and retrieves the CTH and COT [9]. Optical cloud recognition algorithms (OCRAs) is the official CF retrieval method for TROPOMI, which is used as input for the ROCINN-CRB and ROCINN-CAL method. In summary, the OCRA/ROCINN-CAL cloud parameter retrieval scheme makes fewer assumptions, resulting in higher accuracy when validated against other cloud products, and in this study, ROCINN refers to ROCINN-CAL [9].

Compared to the O_2-A band retrieval method used by ROCINN, the O_4 band-based cloud retrieval method can directly extract the O_4 slant column density (SCD) as a feature from observed spectra, eliminating the need for spectral optimization of ROCINN. Furthermore, the O_4 band which is at 460–490 nm exhibits lower surface vegetation reflectance than the O_2-A band, making it less susceptible to seasonal changes in surface vegetation [24], [25], [26]. Additionally, for instruments lacking the O_2-A observation channel, the O_4-band-based cloud parameter retrieval method can provide valuable cloud information. For example, OMCLDO2 is the official algorithm based on the O_4 band applied to OMI observations, which simultaneously retrieves the equivalent radiative CF and cloud pressure while assuming Lambertian cloud properties with a fixed albedo of 0.8 [27]. However, research on cloud parameter retrieval methods in the O_4 band remains relatively limited, particularly those aimed at the simultaneous retrieval of COT and CTH [24], [25], [26], [28], [29].

Recently, the application of machine learning in atmospheric remote sensing has gained significant traction [30]. For instance, it is effective to increase computational efficiency by using machine learning models as an approximate substitute for radiative transfer models (RTMs) due to the limited efficiency of radiative transfer simulations, and ROCINN replaces hyperspectral radiative transfer simulations with neural network models [9]. Similarly, Shi et al. [31] proposed a new algorithm to simultaneously retrieve aerosol and water substances, which was combined with an artificial neural network solver to perform the radiative transfer calculation. In addition, machine learning can also directly learn complex nonlinear relationships between satellite observations and physical parameters of the atmosphere. For example, Wang et al. [32] harnessed the random forest method, using cloud mask and cloud phase state information from CALIOP as labeled data, to conduct cloud detection and phase state identification for VIIRS observations. Similarly, Ri et al. [33] achieved cloud-top parameter retrieval using the random forest method with AHI multichannel observation data, using cloud-top products from CALIOP as a matched labeled dataset for training. Additionally, Tana et al. [34] employed machine learning to retrieve cloud microphysical parameters for Himawari-8, particularly within the sun glint region, utilizing Himawari-8 cloud microphysical parameter products outside the sun glint region as labels. However, while machine learning-based retrieval methods excel in specific applications, there are still several limitations. First, the availability of accurate labeled data is limited, as demonstrated in the coarse spatial resolution of CALIOP lidar observations. Additionally, the direct establishment of nonlinear relationships between observations and physical parameters can potentially hinder the transportability and interpretability of machine learning retrieval methods [35]. Nevertheless, transfer learning presents a promising approach to address these limitations of traditional machine learning. For example, a general and interpretable retrieval model can be trained using the data generated by RTM simulations, further enhancing its efficiency by fine-tuning it with a limited amount of active observation data.

In summary, as the spectral resolution of instruments improves, the efficiency of traditional optimal estimation methods rooted in physical models diminishes. In contrast, machine learning-based retrieval schemes demand extensive training data and are hampered by the quality of labeled data, resulting in limited model transferability and interpretability [22], [26], [28], [29], [36]. Additionally, several instruments, such as the environmental trace gas monitoring instrument (EMI), still lack O_2-A channel observations, and studies simultaneously retrieving CTH and COT using the O_4 band are rare. To address these challenges, this article proposes a novel machine-learning approach, cloud retrieval algorithm based on neural networks using the O_4 (CRANN-O4), for the simultaneous retrieval of CTH and COT. CRANN-O4 leverages a fusion of physical RTMs and multisource satellite information. First, the differential optical absorption spectroscopy (DOAS) technique is employed to directly derive the O_4 SCD, circumventing the need for direct spectrum optimization. Second, CRANN-O4 assumes scattering clouds instead of Lambertian clouds for radiative simulations, effectively rectifying errors stemming from the Lambertian cloud assumption. Finally, CRANN-O4 employs a neural network model with a transfer learning strategy that combines a RTM with CALIOP and VIIRS observations.

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The article is structured as follows: In Section II, we introduce the satellite data used in this study. Section III includes a sensitivity analysis of CTH and COT with respect to O_4 SCD and TOA reflectance at 468 nm. This section also elaborates on the physical principles behind the new algorithm and outlines the algorithmic process. Section IV presents the algorithmic results, offering a comparison with the TROPOMI official product. We also validate the new algorithm by comparing it with the official TROPOMI algorithm using CALIOP data. Finally, in Section V, we provide a summary of the study.

II. Data

A. TROPOMI Observations and Official Cloud Product

The Sentinel-5 precursor satellite, launched in 2017, orbits at an altitude of approximately 824 km with an equator-crossing local solar time of 13:30 [20], [37]. TROPOMI is a hyperspectral atmospheric sounding instrument onboard Sentinel-5P, with a swath of approximately 2600 km, enabling global daily monitoring of aerosols and trace gases such as NO_2, SO_2, O_3, and CH_4 [38], [39], [40], [41], [42]. The spatial resolution of TROPOMI outperforms similar instruments, achieving 3.5 × 5.5 km. This high spatial resolution is valuable for detailed monitoring of atmospheric composition. Furthermore, TROPOMI maintains a high spectral resolution, with its O_4-band spectral resolution surpassing 0.5 nm. Officially, TROPOMI provides multiple cloud parameter products, including OCRA for CF and ROCINN-CAL for CTH and COT [9]. Notably, in this study, similar to the ROCINN algorithm, the OCRA CF is used as the model input. For comparative validation of the algorithms, we rely on the CTH and COT derived from the ROCINN cloud product.

B. SNPP VIIRS Cloud Product

The Suomi national polar-orbiting partnership (SNPP) was launched on October 28, 2011, with an orbital altitude of 833 km and an equator-crossing local solar time of approximately 13:25, slightly earlier than Sentinel-5 precursor [43]. To establish a long-term continuous cloud observation product for climate research, NASA has used VIIRS as a continuation of the MODIS observations, providing reliable cloud property products such as COT, similar to those of MODIS [32]. VIIRS is a multispectral instrument onboard SNPP, with a total of 22 bands, including nine visible and near-infrared bands, a total of 12 mid-infrared and far-infrared bands, and one day/night band (DNB) band. In this article, the matching VIIRS COT products are resampled to the TROPOMI observation latitude/longitude grid, which is used as the labeled data of COT in transfer learning and used for algorithm validation.

C. CALIOP Cloud Layer Measurements

Satellite-based active measurements play a crucial role in cloud and aerosol detection [44]. Among these instruments, CALIOP stands out as an active lidar instrument, part of the A-Train constellation, and deployed onboard CALIPSO. CALIPSO orbits at an altitude of approximately 689 km with an equator-crossing local solar time of 13:45. Operating in the 532 and 1064 nm wavelength bands, CALIOP is primarily engineered to capture vertical structural information concerning tropospheric clouds and aerosols [32], [33]. The CTH products generated by CALIOP exhibit exceptional accuracy, making them a primary data source for validating algorithms related to passive instruments [45]. Therefore, in this study, we rely on the matched CALIOP CTH product, which offers a 5-km resolution, as the reference labeled data for CTH from previous research and for algorithm validation.

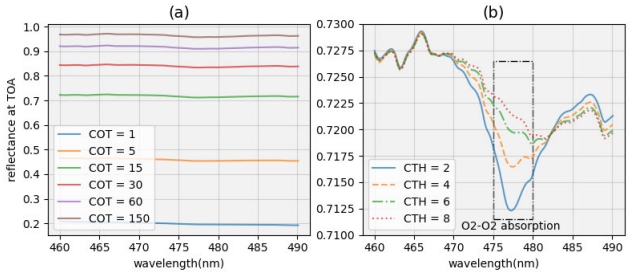


Fig. 1. RTM generated reflectance spectra in different cloud scenarios. (a) Reflectance spectra for different COT. (b) Reflectance spectra for different CTH.

III. Method

A. Sensitivity Study

In the presence of clouds, the backscattered signal observed in the O_4 band by satellites is primarily influenced by COT. As shown in Fig. 1(a), varying the COT leads to significant differences in the observed top-of-atmosphere reflectance, and specifically, the greater the optical thickness is, the higher the observed reflectance. Fig. 1(b) illustrates a prominent absorption feature in the 460–490 nm range, attributed to the collision of oxygen (O_2) molecules and therefore known as O_4 absorption [46]. The absorption cross section of O_4 displays a single-peak distribution, with the peak occurring at 477 nm within the 460–490 nm range. This absorption phenomenon is primarily associated with the concentration of atmospheric O_2, which is uniformly mixed throughout the atmosphere [47]. Consequently, the concentration distribution of O_4 in the atmosphere can be approximated as uniformly mixed. During the transmission of solar radiation, cloud cover significantly affects the path of solar radiation transmission, resulting in variations in the observed degree of O_4 absorption when compared to clear sky conditions. As shown in Fig. 1(b), when clouds are positioned at higher altitudes, solar radiation is reflected at the cloud-top, leading to a shorter path for solar radiation transmission and a lower degree of O_4 absorption along this path [26].

To quantitatively characterize the extent of O_4 absorption, the O_4 SCD along the radiative transfer path was calculated in this study by fitting the backscattered radiance using the DOAS method [48]. The DOAS method takes into account that the observed spectra can be decomposed into two primary components: slow-varying and fast-varying components. Specifically, factors such as Rayleigh scattering, Mie scattering, and surface reflection contribute to spectral variations in reflectance that change slowly with wavelength. Gas molecule

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absorption can also be divided into two components based on wavelength behavior: a fast-varying part and a slow-varying part, as denoted in the following equation:

$$\sigma_i(\lambda) = \sigma_i'(\lambda) + \sigma_i^S(\lambda). \tag{1}$$

In (1), $\sigma_i(\lambda)$ represents the absorption cross section of a gas molecule, $\sigma_i'(\lambda)$ signifies the portion that exhibits rapid wavelength-dependent changes, and $\sigma_i^S(\lambda)$ represents the segment that changes gradually with wavelength. In addition to gas absorption, the ring effect resulting from Raman scattering is likewise categorized into slow and fast variations, which can be expressed in terms of the absorption cross section

$$-\ln R(\lambda) = \sum_i \sigma_i'(\lambda, T)\mathrm{SCD}_i + \sum_{k=0} b_k \lambda^k. \tag{2}$$

As illustrated in (2), the DOAS method substitutes the slow-varying part of the signal with a low-order polynomial represented as $\sum_{k=0} b_k \lambda^k$, while the remaining fast-varying component $\sum_i \sigma_i'(\lambda, T)\mathrm{SCD}_i$ is attributed to gas absorption, $R(\lambda)$ denotes the top-of-atmosphere reflectance. This separation process is accomplished by leveraging gas absorption cross sections measured in laboratory settings, enabling the isolation of the SCD of the target gas from the fast-varying components.

To streamline the radiative transfer modeling, this article employs the IPA assumption. This assumption posits that the reflectance of an individual pixel can be considered a weighted combination of the reflectance of the cloudy subpixel and the clear sky subpixel

$$R(\lambda) = (1 - \mathrm{cf}) \times R^{\mathrm{clear}}(\lambda, A_{\mathrm{surface}}, H_{\mathrm{surface}}) + \mathrm{cf} \times R^{\mathrm{cloud}}(\lambda, \tau_{\mathrm{cloud}}, H_{\mathrm{cloud}}). \tag{3}$$

In (3), $R(\lambda)$ denotes the top-of-atmosphere reflectance, cf denotes the equivalent radiative CF, R^{clear} and R^{cloud} denote the top-of-atmosphere reflectance for clear subpixel and cloudy subpixel, respectively, A_{surface} denotes the surface albedo, H_{surface} denotes the surface elevation, τ_{cloud} denotes the COT, and H_{cloud} denotes the CTH. As an illustrative example, the top-of-atmosphere (TOA) reflectance at 468 nm that experiences weak absorption by the atmosphere is considered, as shown in Fig. 1(b). As depicted in Fig. 2, with fixed surface parameters, simulations using the SCIATRAN model demonstrate that the top-of-atmosphere reflectance varies primarily according to CF and COT.

Similarly, the calculated O_4 SCD can be approximated as a combination of contributions from the clear-sky subpixel and the cloudy subpixel, as indicated in (4). To investigate the relationship between the O_4 SCD, CF, and CTH, backscattering spectra for various CF, CTH, and surface albedos were simulated while maintaining a constant COT of 20. As depicted in Fig. 3, it is obvious that the O_4 SCD is primarily influenced by the CF and CTH across different surface albedo scenarios. Notably, within the same CF scenario, an increase in CTH corresponds to a decrease in O_4 SCD

$$\mathrm{SCD}_{O_4} = (1 - \mathrm{cf}) \times \mathrm{SCD}_{O_4}^{\mathrm{clear}}(\theta, \phi, A_{\mathrm{surface}}, H_{\mathrm{surface}}) + \mathrm{cf} \times \mathrm{SCD}_{O_4}^{\mathrm{cloud}}(\theta, \phi, \tau_{\mathrm{cloud}}, H_{\mathrm{cloud}}). \tag{4}$$

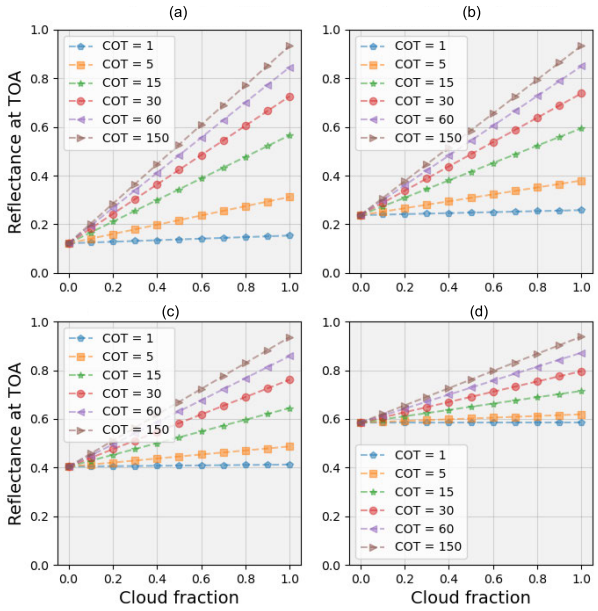


Fig. 2. RTM generated 468 nm TOA reflectance in different cloud scenarios and surface albedos with a CTH of 4 km. (a) Surface albedo = 0.05. (b) Surface albedo = 0.2. (c) Surface albedo = 0.4. (d) Surface albedo = 0.6.

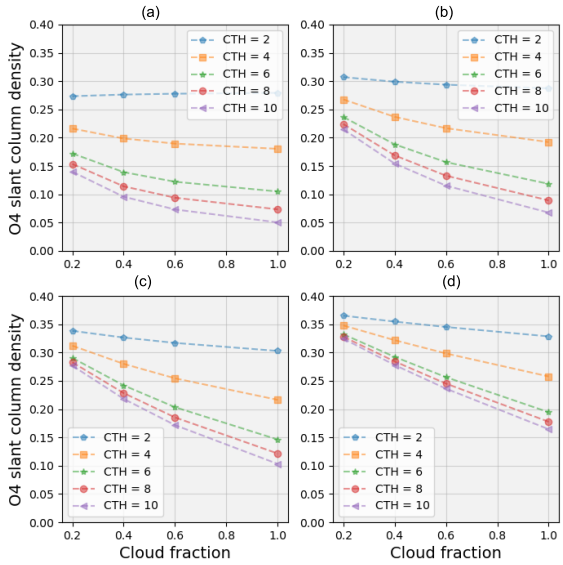


Fig. 3. Calculated O_4 SCD from the simulated spectrum for different cloud scenarios and surface albedos with COT at 20. (a) Surface albedo = 0.05. (b) Surface albedo = 0.2. (c) Surface albedo = 0.4. (d) Surface albedo = 0.6.

In (4), SCD_{O_4} denotes the O_4 SCD, θ, ϕ denotes the viewing geometry, and other parameters are the same as above.

Using the TROPOMI observations from March 1, 2020, as a case study, Fig. 4 presents grayscale radiance images at 468 nm from TROPOMI, ROCINN-CAL COT, O_4 SCD, and ROCINN-CAL CTH for various surface type scenarios. These surface types include snow and ice areas (column a), wilderness (column b), and ocean (columns c and d). Upon examining the figure, several noteworthy findings emerge. First, it becomes evident that the overall spatial distribution of COT and radiance values align closely. The alignment indicates that COT exerts a significant influence on the

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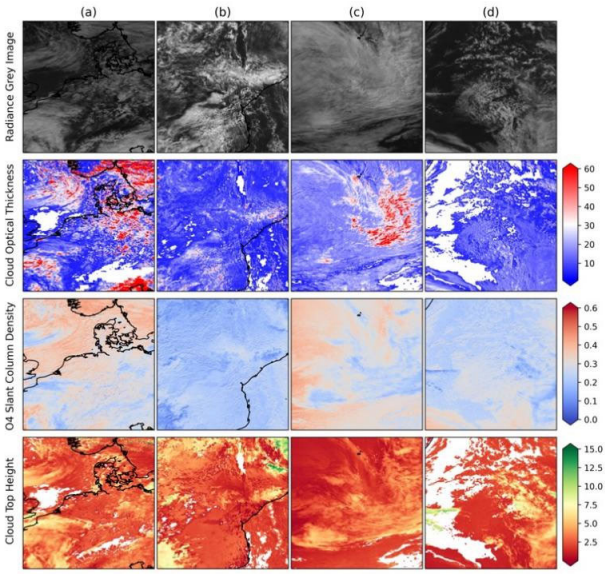


Fig. 4. Example of grayscale radiance image, COT from ROCINN, O_4 SCD and CTH from ROCINN spatial distribution for various surface type scenarios. (a)–(d) Show the grayscale radiance image, CF, O_4 SCD, and CTH, respectively.

observed radiance. Additionally, there is a high level of spatial consistency between the O_4 SCD and the distribution of CTH. Based on the sensitivity analysis described above, it has been determined that CTH and COT are primarily influenced by factors such as TOA reflectance and O_4 SCD when surface parameters and observation geometry are determined.

B. Construction of the Neural Network-Based Retrieval Model (CRANN-O4)

To address the challenges of slow retrieval efficiency, generally low cloud height and generally high COT were observed in the traditional optimal estimation-based ROCINN-CAL method when compared to the VIIRS cloud product. This article developed a machine-learning-based cloud parameter retrieval algorithm (CRANN-O4) based on radiative transfer principles and multisource satellite data using the O_4 band from a hyperspectral instrument. A transfer-learning strategy was adopted to combine the dataset from radiative transfer simulation and multisource satellites. As discussed in the introduction section, several challenges arise in constructing this algorithm. First, direct training of machine learning models using observational data lacks physical explainability and universality. Second, the labeled dataset for training is inadequate, and the sparse observations from CALIOP pose a challenge in obtaining sufficient labeled data for training. In contrast, a transfer-learning strategy offers an effective solution by requiring a smaller amount of labeled data while retaining the physical model basis and introducing observations to fine-tune the model.

As depicted in Fig. 5, the process begins with the random simulation of spectra under various cloud scenarios and surface parameters using the SCIATRAN RTM [49], [50]. Based on this spectral database, the O_4 SCD is calculated using the DOAS method, forming the labeled dataset for training. Input features include observation geometry, surface conditions,

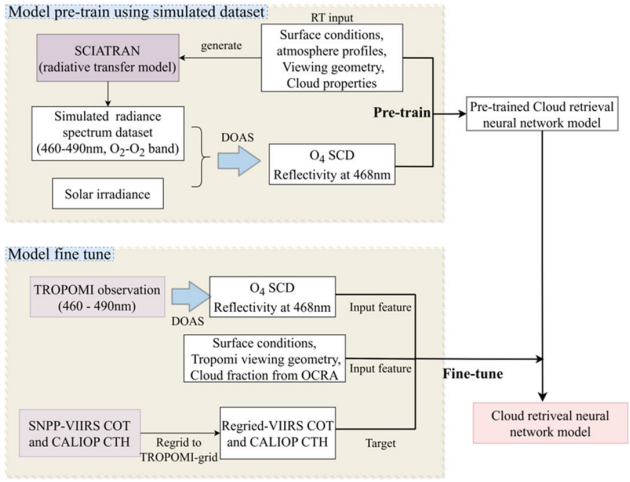


Fig. 5. Flowchart of the CRANN-O4 model, consisting mainly of two parts: pretraining using the simulated dataset generated by the RTM and model fine-tuning using CALIOP CTH and SNPP-VIIRS COT.

O_4 SCD, reflectance at 468 nm, and CF, while the outputs encompass CTH and COT. To establish the relationship between the input and output data, a neural network model with four hidden layers is constructed and trained using the dataset derived from the RTM. The neural network model employed in this study features four hidden layers, with the number of neurons per layer set to 128, 128, 256, and 256, respectively. The activation function selected for this architecture is Leaky_ReLU. During the pretraining phase, we utilized a dataset comprising approximately 1.3 million samples generated randomly using a RTM. The dataset was divided into training, testing, and validation subsets at a ratio of 7:2:1. The training pattern was adopted to optimize the weights and biases of each neuron iteratively. In each epoch, the testing pattern was used to monitor the performance of the neural network model to avoid overfitting during the data training. In the fine-tuning phase, considering the substantial volume of hyperspectral instrument data and the varying transit times between CALIOP and TROPOMI, we opted to use the data from the first of each month in 2020 for training, testing, and validation purposes. Subsequently, CALOP and VIIRS cloud product data are introduced to fine-tune the pretrained neural network model generated based on the RTM. During the model adjustment phase, the weight parameters of the first two hidden layers remain fixed, with only the last two layers being adjusted. In this study, CALIOP-observed CTH and VIIRS-observed COT serve as labeled data for model fine-tuning.

IV. Results and Discussion

A. Retrieval Results From TROPOMI Observations

CRANN-O4 has been applied to TROPOMI observations for the retrieval of global COT and CTH to validate the effectiveness of CRANN-O4. Fig. 6 illustrates the COT and CTH on March 1, 2020, derived from ROCINN and CRANN-O4, respectively. Upon examining the retrieval results, it becomes apparent that the spatial distributions of cloud parameters derived from CRANN-O4 align consistently

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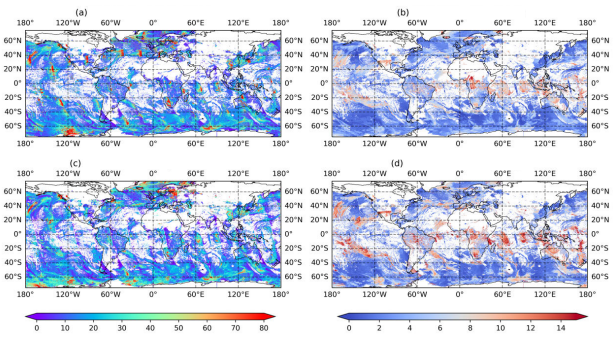


Fig. 6. Comparison of CTH and COT results from ROCINN and the new algorithm for March 1, 2020. (a) ROCINN COT. (b) ROCINN CTH. (c) CRANN-O4 COT. (d) CRANN-O4 CTH.

on a global scale. Furthermore, when comparing the CRANN-O4-derived COT with that of ROCINN, it is evident that the COTs derived from CRANN-O4 are overall lower. Additionally, the CTH derived from CRANN-O4 is higher than that derived from ROCINN.

B. Case Study of CTH and COT Retrieval

To assess the performance disparity between ROCINN and CRANN-O4 in CTH and COT retrieval, we conducted a comparative analysis of spatial distributions. Fig. 7 presents the spatial distribution of various parameters, including TROPOMI-observed radiance at 468 nm, retrieved O_4 SCD, COT, and CTH from ROCINN, COT and CTH from CRANN-O4, and COT and CTH from SNPP-VIIRS. Taking COT as an example, Fig. 7(b) highlights a region with notably higher COT values. This region is evident in ROCINN's results but is absent in VIIRS and CRANN-O4's COT retrievals. Additionally, it is noticeable that ROCINN generally produces higher COT values compared to VIIRS, a deviation that our algorithm successfully mitigates. Regarding CTH results, when compared to VIIRS official CTH, ROCINN tends to underestimate CTH in the northern region of Australia, and our algorithm has significantly improved this issue. This analysis aims to highlight the improvements made by CRANN-O4 in addressing ROCINN's challenges of generally low cloud height values and high COT values by comparing the cloud properties of the Tropical Cyclone Esther from ROCINN and CRANN-O4 for March 1, 2020.

As illustrated in Fig. 8, taking Typhoon Maysak observed on September 1, 2020, as an example, we can observe a distinct, larger region with high COT values in Fig. 8(b), which is not present in the COT results obtained from either the VIIRS official product or CRANN-O4. Regarding the CTH retrieval results, ROCINN exhibits significantly lower estimates in the East China Sea region and the Philippine Sea region when compared to the cloud height results obtained from Himawari-8. On the other hand, CRANN-O4 significantly ameliorates this issue, with estimates more closely aligned with CTHs derived from SNPP-VIIRS.

In Fig. 9, we present the validation results for COT and CTH derived from CRANN-O4 and ROCINN, comparing them to the SNPP-VIIRS COT and CTH results. Compared with the SNPP-VIIRS cloud product, the RMSEs of COT and CTH derived by CRANN-O4 are approximately 16.29 and

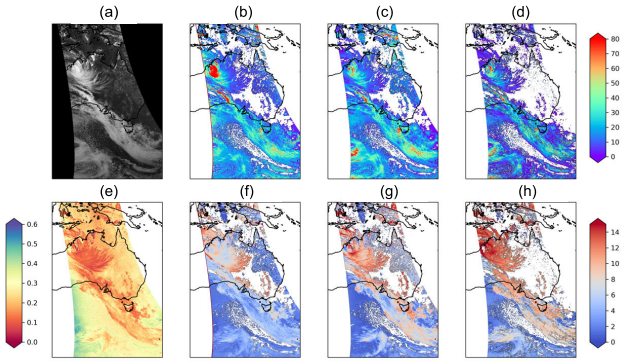


Fig. 7. Comparison of COT and CTH from ROCINN and CRANN-O4 of Tropical Cyclone Esther on March 1, 2020. (a) Grayscale radiance image from TROPOMI observations. (b) TROPOMI official COT. (c) CRANN-O4 COT. (d) SNPP-VIIRS official COT. (e) Calculated O_4 SCD from TROPOMI observations. (f) TROPOMI official CTH. (g) CRANN-O4 CTH. (h) SNPP-VIIRS official CTH.

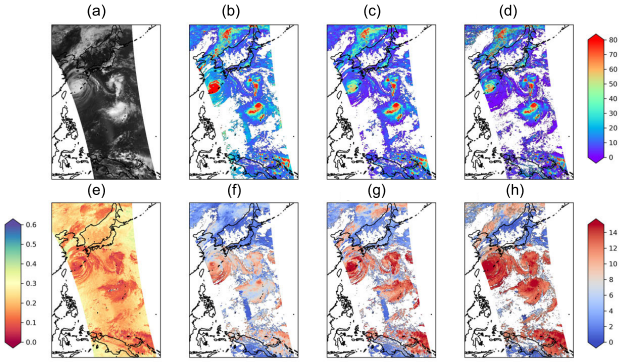


Fig. 8. Comparison of COT and CTH from the ROCINN and CRANN-O4 14 950th orbit on September 1, 2020. (a) Grayscale radiance image from TROPOMI observations. (b) TROPOMI official COT. (c) CRANN-O4 COT. (d) SNPP-VIIRS official COT. (e) Calculated O_4 SCD from TROPOMI observations. (f) TROPOMI official CTH. (g) CRANN-O4 CTH. (h) SNPP-VIIRS official CTH.

2.35 km, respectively, in comparison to that of the TROPOMI cloud product with RMSEs of 21.29 and 2.97 km for the 12 340th orbit on March 1, respectively. Similarly, for the 14 950th orbit on September 1, the RMSEs of COT and CTH derived by CRANN-O4 are approximately 15.55 and 2.31 km, respectively, in comparison to that of the TROPOMI cloud product with RMSEs of 20.50 and 3.03 km, respectively. On average, for the 12 340th and 14 950th orbits, CRANN-O4 achieved an RMSE of 15.88 for COT and 2.33 km for CTH, whereas the official TROPOMI product had higher RMSE values of 20.85 for COT and 3.00 km for CTH. These findings highlight that CRANN-O4 provides COT and CTH values that closely align with the VIIRS cloud properties, representing an improvement over the higher RMSE observed with ROCINN. For the COT retrieval, part of the improvement may be because the surface albedo in the O_4 band is generally lower than that in the O_2-A band. In addition, a transfer learning approach was adopted, and the model was fine-tuned using the VIIRS COT, avoiding bias from the radiative transfer calculations due to the IPA assumption.

C. Validation of CTH With CALIOP Measurements

An independent dataset was selected to validate the effectiveness of our CTH retrieval algorithm. Fig. 10(a) provides a

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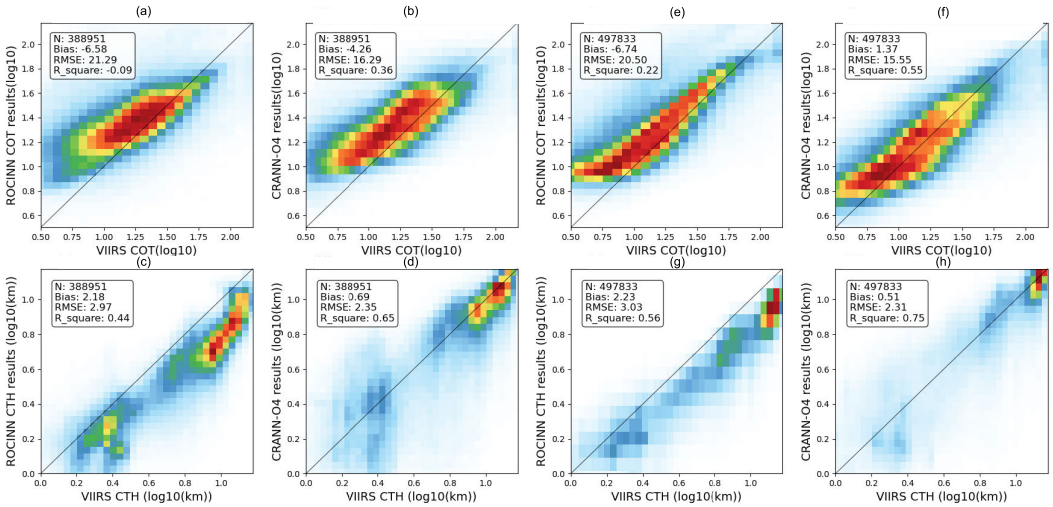


Fig. 9. Comparison of COT and CTH derived from ROCINN and CRANN-O4 with SNPP-VIIRS. (a) and (b) Scatter plot of the ROCINN COT and CRANN-O4 COT with VIIRS COT of the 12 340th orbit on March 1. (c) and (d) Scatter plot of the ROCINN CTH and CRANN-O4 CTH with VIIRS CTH of the 12 340th orbit on March 1. (e) and (f) Scatter plot of the ROCINN COT and CRANN-O4 COT with VIIRS COT of the 14 950th orbit on September 1. (g) and (h) Scatter plot of the ROCINN CTH and CRANN-O4 CTH with VIIRS CTH of 14 950th orbit on September 1.

comparison of CTH from CALIOP measurements, CRANN-O4 results, and ROCINN results on March 1, 2020. The figure clearly demonstrates that CRANN-O4-derived CTH exhibits a consistent trend with ROCINN-derived CTH. Additionally, Fig. 10(d) and (e) showcases scatter validation plots depicting the CTH derived from ROCINN and CRANN-O4 against CALIOP-derived CTH. It is evident that CTH from CRANN-O4, particularly in the high cloud layer, closely aligns with the 1:1 reference line compared to ROCINN. Quantitatively, in comparison to CALIOP CTH, our algorithm achieves an RMSE of 1.9 km, outperforming ROCINN, which yields an RMSE of 3.4 km. Furthermore, the coefficient of determination (R^2) for our algorithm is 0.87, significantly surpassing ROCINN's R^2 of 0.59.

Similarly, as shown in Fig. 11, the validation dataset for September 1, 2020, was chosen for the validation of the CRANN-O4's performance. Fig. 11(a) provides a comparison of CTH from CALIOP measurements and the results of CRANN-O4 and ROCINN for September 1, 2020. Quantitatively, in comparison to CALIOP CTH, our algorithm achieves an RMSE of 1.9 km, outperforming ROCINN, which yields an RMSE of 3.2 km. Furthermore, the R^2 value for CRANN-O4 is 0.83, significantly surpassing ROCINN's R^2 of 0.55.

Finally, we conducted a comprehensive validation of CTH retrieval results from various algorithms with CALIOP measurements for March 1 and September 1, 2020. Fig. 12(a) and (b) depicts the results of the comparison between the ROCINN, CRANN-O4, and CALIOP measurements on March 1, 2020. It is evident from Fig. 12 that ROCINN retrieval results tend to underestimate cloud heights compared to CALIOP measurements, a discrepancy that is notably improved by CRANN-O4. Specifically, ROCINN exhibits an RMSE of 2.9 km and an R^2 value of 0.63. In contrast, CRANN-O4 achieves an RMSE of 2.2 km and an R^2 of 0.78 for the March 1 comparison, indicating its superiority over ROCINN. Similarly, the comparison validation conducted for September 1, 2020, yields consistent results. The ROCINN demonstrates an RMSE of 2.7 km and an R^2 of

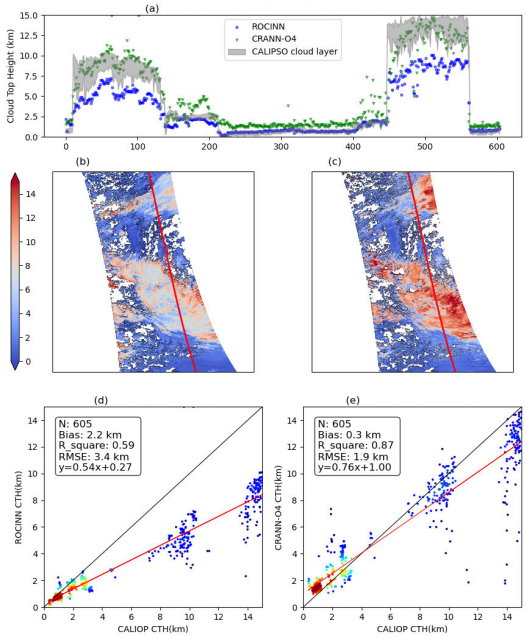


Fig. 10. Point-by-point comparison of CTH results from the CRANN-O4 and ROCINN methods versus CALIOP data from 23:17:57 UTC time on February 29 to 00:10:17 UTC time on March 1, 2020. (a) Scatter plot of the CALIOP CTH measurements and the CRANN-O4, ROCINN CTH. (b) and (c) ROCINN and CRANN-O4 CTH results and matched CALIOP observation trajectory. (d) and (e) Scatter density plots of the CALIOP, CRANN-O4, and ROCINN CTH.

0.62, whereas our algorithm achieves an RMSE of 2.3 km and an R^2 of 0.73. On average, CRANN-O4 achieved an RMSE of 2.2 km, whereas the TROPOMI official product had higher RMSE values of 2.7 km. The results of global comparisons throughout different seasons affirm the robustness of our algorithm across varying conditions, consistently outperforming ROCINN according to different evaluation metrics.

To investigate the performance of the transfer-learning strategy used in this study, we further made comparisons of retrieved CTH and COT with another two neural networks using the same structure but different data sources,

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TABLE I
PERFORMANCE OF DIFFERENT TRAINING STRATEGY

	pre-training with fine tuning	pre-training alone from RT model	pre-training alone from observation data
RMSE (CTH)	1.9km	4.5km	3.1km
R^2 (CTH)	0.87	0.3	0.68
Mean Bias (CTH)	0.3km	2.6km	3.1km
RMSE (COT)	15.55	25.44	17.83
R^2 (COT)	0.55	-0.21	0.41
Mean Bias (COT)	1.37	-7.00	3.82

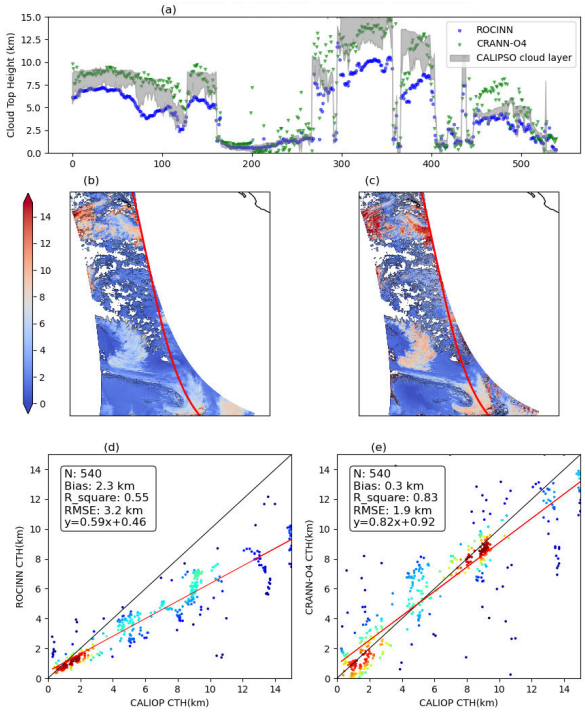


Fig. 11. Point-by-point comparison of CTH results from the CRANN-O4 and ROCINN methods versus CALIOP data from 23:37:10 UTC time on September 1 to 00:29:30 UTC time on September 2, 2020. (a) Scatter plot of the CALIOP CTH measurements and the CRANN-O4, ROCINN CTH. (b) and (c) ROCINN and CRANN-O4 CTH results and matched CALIOP observation trajectory. (d) and (e) Scatter density plots of the CALIOP, CRANN-O4, and ROCINN CTH.

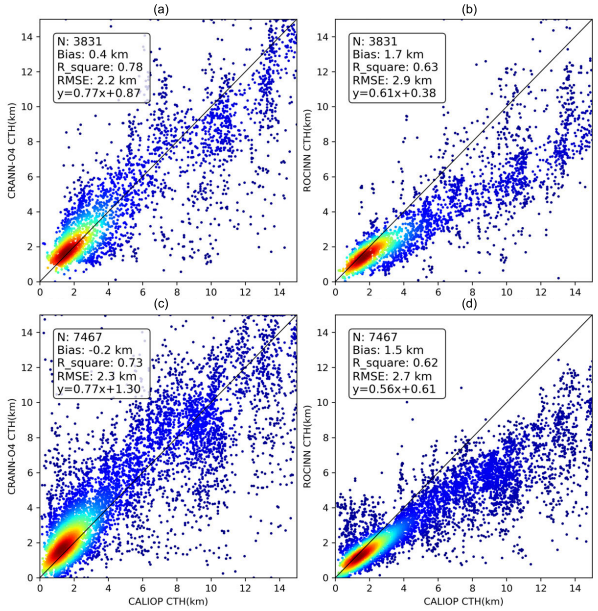


Fig. 12. Point-by-point comparison of CTH results obtained from different retrieval algorithms versus CALIOP measurements. (a) and (c) Scatter plot of the CALIOP CTH measurements and the CRANN-O4 CTH on March 1 and September 1, 2020, respectively. (b) and (d) Scatter plot of the CALIOP CTH measurements and the ROCINN CTH on March 1 and September 1, 2020, respectively.

i.e., pretraining from RTM alone and pretraining from the CALIOP/VIIRS observation data alone. The specific validation results of CTH and COT on September 1, 2020 between these three strategies are shown in Table I. Results demonstrate that the neural network model using the transfer-learning strategy, i.e., CRANN-O4 algorithm, with a combination of training data generated by the RTM and the observation data, is the most effective in the retrieval of CTH and COT with the lowest RMSE and the highest accuracy. In contrast, the model trained using only the data generated by the RTM has both a larger systematic bias and a lower consistency with the observation data, no matter for the retrieval of CTH or COT. Such conditions are also shown when using the neural network model trained only from the observation data, which is mainly due to the limited reliable training samples that can be provided at present. Meanwhile, this strategy of directly applying observation data for training makes it difficult to migrate the algorithmic model between different instruments due to the lack of physical foundation. In general, the transfer-learning strategy with pretraining data from a combination of RTM and observation data show the best performance in the simultaneous retrieval of COT and CTH from the O_4 bands.

V. CONCLUSION

In this article, a transfer learning-based hyperspectral cloud parameter retrieval algorithm using the O_4 band is proposed for retrieving CTH and COT. We initiate the study by conducting a sensitivity analysis using radiative transfer forward simulations. The sensitivity analysis demonstrates that O_4 SCD is predominantly influenced by CTH, CF, and COT, while TOA reflectance at 468 nm is primarily determined by CF and COT.

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Furthermore, we conduct separate comparisons of the observed radiance data from TROPOMI, O_4 SCD, and cloud products. The comparison shows that the COT and the observed radiance have the similar spatial distribution, while the CTH and O_4 SCD have the similar spatial distribution. Based on the outcomes of these sensitivity analyses, we determined that the O_4 SCD and 468 nm reflectance are effective indicators for CTH and COT retrieval. In this article, we develop a deep neural network model based on the O_4 absorption band for CTH and COT retrieval for hyperspectral instruments. Compared to the traditional cloud retrieval method based on the O_2-A band, such as ROCINN, the present method circumvents the need for direct spectral optimal estimation, resulting in improved inversion efficiency. Given the limited availability of labeled data for training, this article employs a transfer learning strategy. First, the model is pretrained based on the spectral dataset randomly generated by the SCIATRAN RTM. Then, the weight parameters of the first two hidden layers remain fixed, and the model is adjusted with matched cloud height data from CALIOP and matched COT data from VIIRS. This transfer learning strategy preserves the feature extraction capabilities of the neural network model trained with data generated by the RTM. Simultaneously, it integrates multisource satellite information to mitigate retrieval errors caused by assumptions in the RTM.

We applied the CRANN-O4 to TROPOMI observations to retrieve CTH and COT and subsequently conducted independent validation of the retrieval results. Two scenarios from Tropical Cyclone Esther and Typhoon Maysak are selected in the article, and the CTH and COT derived from ROCINN and CRANN-O4 are given. Additionally, the SNPP-VIIRS official cloud product is also given as a reference. The retrieval results demonstrate that our algorithms effectively mitigate the generally low bias in CTH and the generally high bias in COT. Furthermore, we conduct quantitative validation of the results obtained by both algorithms. Compared with the SNPP-VIIRS cloud product, the RMSEs of COT and CTH derived by CRANN-O4 are 15.88 and 2.33 km, respectively, in comparison to that of the TROPOMI cloud product with RMSEs of 20.85 and 3.00 km, respectively. Compared with the CALIOP CTH measurements, the RMSE for the CTH results improves from 2.7 km with ROCINN to 2.2 km with CRANN-O4, while the R^2 values increase from 0.63 to 0.75.

In summary, the neural network model based on the transfer learning strategy proposed in this article demonstrates outstanding performance in retrieving CTH and COT. As the spectral resolution of hyperspectral instruments continues to improve, the algorithm using the O_4 absorption band introduced in this article becomes increasingly computationally efficient compared to the ROCINN method, which necessitates optimal spectral estimation. Furthermore, the retrieval model, which relies on the RTM and multisource satellite information, leverages a substantial volume of randomly generated spectral data during pretraining, while the model fine-tuning phase requires fewer labeled observations. This makes it more convenient to adapt the model to other instruments, such as the ozone monitoring suite (OMS) onboard the FY-3F and the EMI onboard the GaoFen-5. However, similar to ROCINN, the algorithm proposed in this article utilizes the CF from OCAR as an input feature, necessitating the generation of CF products in cloud product production. The synchronization of the CF, CTH, and COT retrieval is an area that requires ongoing improvement in the future. Note that we adopt the water cloud assumption in this study, which is the same as the official TROPOMI algorithms of ROCINN due to the lack of bands for cloud phase discrimination. Sentinel-5 is scheduled to be launched in 2025 and has similar observational performance to TROPOMI, but with the addition of 1.6 μm observations. Therefore, we plan to utilize 1.6 μm observations to discriminate cloud phase states in our future study.

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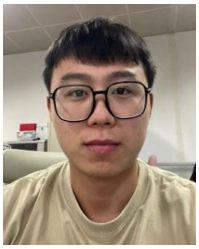
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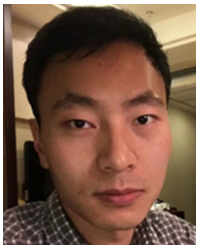
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